

Solution Requirements

AI-Assisted Smart Budget Planner

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Registration, login, logout, JWT/session-cookie authentication and scrypt password hashing.	High
2	Planner Functions	Home, Party and Jewelry planners accept user preferences and budget and return itemized plans.	High
3	Budget Processing	The shared packing engine keeps estimated total $\leq$ requested budget.	High
4	External Interfaces	Generate search links for Amazon, Flipkart, IKEA, Swiggy and OYO; optional Gemini API integration.	Medium
5	Image Upload	Jewelry planner accepts optional JPEG/PNG/WebP outfit image up to 3 MB with signature validation.	Medium
6	History / Reporting	Save plans per user and retrieve history and plan details.	High
7	Testing	Automated tests cover registration, login, planners, budget caps and image validation.	High

Step 2: Non-Functional Requirements (NFR)

Category	Requirement
Security	Salted scrypt password hashing; HTTP-only SameSite=Lax cookies; JWT support.
Reliability	Graceful Gemini fallback when no API key is configured.
Usability	Responsive static HTML/CSS/JavaScript interface.
Maintainability	Typed Pydantic models and shared planner logic.
Performance	Deterministic packing avoids expensive optimization for the sample catalog.
Production Readiness	Use HTTPS, COOKIE_SECURE=true, rate limiting and persistent DB/backups in production.